Министерство образования Республики Беларусь Учреждение образования
«Брестский государственный технический университет»
Кафедра ИИТ

Лабораторная работа №3
По дисциплине: «Обработка изображений в интеллектуальных системах»
По теме: «**Обучение детекторов объектов**»

**Выполнил:**
студент 4 курса
группы ИИ-25
Гродзинский А.Д.
**Проверил:**
Андренко К.В.

Брест 2026

**Цель:** осуществлять обучение нейросетевого детектора для решения задачи обнаружения заданных объектов

| | | |
|---|---|---|
| 4 | YOLO11s | **Коты:** https://universe.roboflow.com/mohamed-traore-2ekkp/cats-n9b87/dataset/3 |

## Результаты работы сети:

```
Creating new Ultralytics Settings v0.0.8 file
View Ultralytics Settings with 'yolo settings' or at 'C:\Users\cimer\AppData\Roaming\Ultralytics\settings.json'
Update Settings with 'yolo settings key=value', i.e. 'yolo settings runs_dir=path/to/dir'. For help see https://docs.ultralytics.com/usage/settings
Downloading https://github.com/ultralytics/assets/releases/download/v8.4.0/yolo…: 100% ━━━━━━━━━━━━ 18.4MB 7.6MB/s 2.4s
Ultralytics 8.4.211  Python-3.13.3 torch-2.15.0+cpu CPU (11th Gen Intel Core i5-11400H @ 2.70GHz)
engine\trainer: agnostic_nms=False, amp=True, angle=1.0, augment=False, auto_augment=randaugment, batch=8, bgr=0.0, box=7.5, cache=False, cfg=None, channels
_last=None, classes=None, close_mosaic=10, cls=0.5, cls_pw=0.0, cls_remap=True, compile=False, conf=None, copy_paste=0.0, copy_paste_mode=flip, cos_lr=False
, cutmix=0.0, data=C:\Lesha\4курс задания\ОИВИС\lab3\dataset\data.yaml, degrees=0.0, deterministic=True, device=cpu, dfl=1.5, dgrad=0.5, dis=6.0, distill_mo
del=None, dlam=1.0, dlog=1.0, dnn=False, dropout=0.0, dynamic=False, embed=None, epochs=5, erasing=0.4, exist_ok=True, fliplr=0.5, flipud=0.0, format=torchs
cript, fraction=1.0, freeze=None, hsv_h=0.015, hsv_s=0.7, hsv_v=0.4, imgsz=640, iou=None, kobj=1.0, line_width=None, lr0=0.01, lrf=0.01, mask_ratio=4, max_d
et=300, mixup=0.0, mode=train, model=yolo11s.pt, momentum=0.937, mosaic=1.0, multi_scale=0.0, name=train, nbs=64, nms=None, opset=None, optimize=False, opti
mizer=auto, overlap_mask=True, patience=100, perspective=0.0, plots=True, pose=12.0, pretrained=True, profile=False, project=C:\Lesha\4курс задания\ОИВИС\la
b3\runs, quantize=None, rect=False, resume=False, retina_masks=False, rle=1.0, save=True, save_conf=False, save_crop=False, save_dir=C:\Lesha\4курс задания\
ОИВИС\lab3\runs\train, save_frames=False, save_json=False, save_period=-1, save_txt=False, scale=0.5, seed=0, shear=0.0, show=False, show_boxes=True, show_c
onf=True, show_labels=True, simplify=True, single_cls=False, source=None, split=val, stream_buffer=False, task=detect, tile=False, time=None, tracker=trackt
rack.yaml, translate=0.1, val=True, verbose=True, vid_stride=1, visualize=False, warmup_bias_lr=0.1, warmup_epochs=3.0, warmup_momentum=0.8, weight_decay=0.
0005, workers=0, workspace=None
Downloading https://ultralytics.com/assets/Arial.ttf to 'C:\Users\cimer\AppDat…: 100% ━━━━━━━━━━━━ 755.1KB 2.4MB/s 0.3s
Overriding model.yaml nc=80 with nc=1

                   from  n    params  module                                       arguments
  0                  -1  1       928  ultralytics.nn.modules.conv.Conv             [3, 32, 3, 2]
  1                  -1  1     18560  ultralytics.nn.modules.conv.Conv             [32, 64, 3, 2]
  2                  -1  1     26080  ultralytics.nn.modules.block.C3k2            [64, 128, 1, False, 0.25]
  3                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
  4                  -1  1    103360  ultralytics.nn.modules.block.C3k2            [128, 256, 1, False, 0.25]
  5                  -1  1    590336  ultralytics.nn.modules.conv.Conv             [256, 256, 3, 2]
  6                  -1  1    346112  ultralytics.nn.modules.block.C3k2            [256, 256, 1, True]
  7                  -1  1   1180672  ultralytics.nn.modules.conv.Conv             [256, 512, 3, 2]
  8                  -1  1   1380352  ultralytics.nn.modules.block.C3k2            [512, 512, 1, True]
  9                  -1  1    656896  ultralytics.nn.modules.block.SPPF            [512, 512, 5]
 10                  -1  1    990976  ultralytics.nn.modules.block.C2PSA           [512, 512, 1]
 11                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 12             [-1, 6]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 13                  -1  1    443776  ultralytics.nn.modules.block.C3k2            [768, 256, 1, False]
 14                  -1  1         0  torch.nn.modules.upsampling.Upsample         [None, 2, 'nearest']
 15             [-1, 4]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 16                  -1  1    127680  ultralytics.nn.modules.block.C3k2            [512, 128, 1, False]
 17                  -1  1    147712  ultralytics.nn.modules.conv.Conv             [128, 128, 3, 2]
 18            [-1, 13]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 19                  -1  1    345472  ultralytics.nn.modules.block.C3k2            [384, 256, 1, False]
 20                  -1  1    590336  ultralytics.nn.modules.conv.Conv             [256, 256, 3, 2]
 21            [-1, 10]  1         0  ultralytics.nn.modules.conv.Concat           [1]
 22                  -1  1   1511424  ultralytics.nn.modules.block.C3k2            [768, 512, 1, True]
 23        [16, 19, 22]  1    819795  ultralytics.nn.modules.head.Detect           [1, 16, None, [128, 256, 512]]
YOLO11s summary: 181 layers, 9,428,179 parameters, 9,428,163 gradients, 21.7 GFLOPs

Remapped 1/1 cls head rows from pretrained weights by class name
Transferred 499/499 items from pretrained weights
Freezing layer 'model.23.dfl.conv.weight'
WARNING train: Slow image access detected (ping: 0.1±0.0 ms, read: 4.7±1.9 MB/s, size: 28.0 KB). Use local storage instead of remote/mounted storage for bet
ter performance. See https://docs.ultralytics.com/guides/model-training-tips
train: Scanning C:\Lesha\4курс задания\ОИВИС\lab3\dataset\train\labels... 81…: 100% ━━━━━━━━━━━━ 811/811 558.4it/s 1.5s
train: New cache created: C:\Lesha\4курс задания\ОИВИС\lab3\dataset\train\labels.cache
WARNING val: Slow image access detected (ping: 0.1±0.0 ms, read: 8.1±5.8 MB/s, size: 44.0 KB). Use local storage instead of remote/mounted storage for bette
r performance. See https://docs.ultralytics.com/guides/model-training-tips
val: Scanning C:\Lesha\4курс задания\ОИВИС\lab3\dataset\valid\labels... 232 …: 100% ━━━━━━━━━━━━ 232/232 576.9it/s 0.4s
val: New cache created: C:\Lesha\4курс задания\ОИВИС\lab3\dataset\valid\labels.cache
optimizer: 'optimizer=auto' found, ignoring 'lr0=0.01' and determining best 'optimizer' and 'lr0' automatically...
optimizer: AdamW(lr=0.002, momentum=0.9) with parameter groups 81 weight(decay=0.0), 88 weight(decay=0.0005), 87 bias(decay=0.0)
Plotting labels to C:\Lesha\4курс задания\ОИВИС\lab3\runs\train\labels.jpg...
Using 811 train, 232 val images for fraction=1.0 at imgsz=640
Using 0 dataloader workers
Logging results to C:\Lesha\4курс задания\ОИВИС\lab3\runs\train
Starting training for 5 epochs...
Closing dataloader mosaic
```

```
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        1/5         0G      1.041     0.9584       1.77          3        640: 100% ━━━━━━━━━━━━ 102/102 3.6s/it 6:10
                   Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 15/15 1.9s/it 27.9s
                     all        232        235      0.694      0.651      0.692      0.321

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        2/5         0G      1.457      1.086      2.131          3        640: 100% ━━━━━━━━━━━━ 102/102 3.4s/it 5:43
                   Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 15/15 1.9s/it 27.9s
                     all        232        235        0.7      0.732      0.755      0.289

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        3/5         0G      1.379     0.9811      2.067          3        640: 100% ━━━━━━━━━━━━ 102/102 3.4s/it 5:42
                   Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 15/15 1.9s/it 27.8s
                     all        232        235      0.676      0.711      0.723      0.334

      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        4/5         0G      1.314     0.8552       2.01          3        640: 100% ━━━━━━━━━━━━ 102/102 3.4s/it 5:43
                   Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 15/15 1.9s/it 27.9s
                     all        232        235       0.78      0.791      0.853      0.406

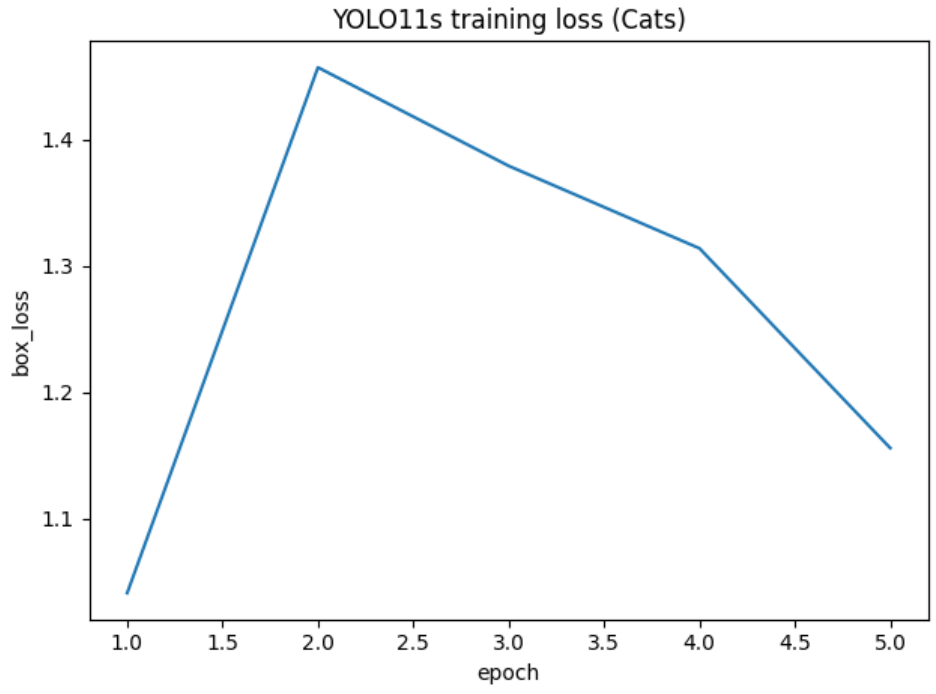
      Epoch    GPU_mem   box_loss   cls_loss   dfl_loss  Instances       Size
        5/5         0G      1.156      0.791      1.826          3        640: 100% ━━━━━━━━━━━━ 102/102 3.4s/it 5:43
                   Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 15/15 1.9s/it 27.8s
                     all        232        235      0.918      0.911      0.945      0.567

5 epochs completed in 0.525 hours.
Optimizer stripped from C:\Lesha\4курс задания\ОИВИС\lab3\runs\train\weights\last.pt, 19.2MB
Optimizer stripped from C:\Lesha\4курс задания\ОИВИС\lab3\runs\train\weights\best.pt, 19.2MB

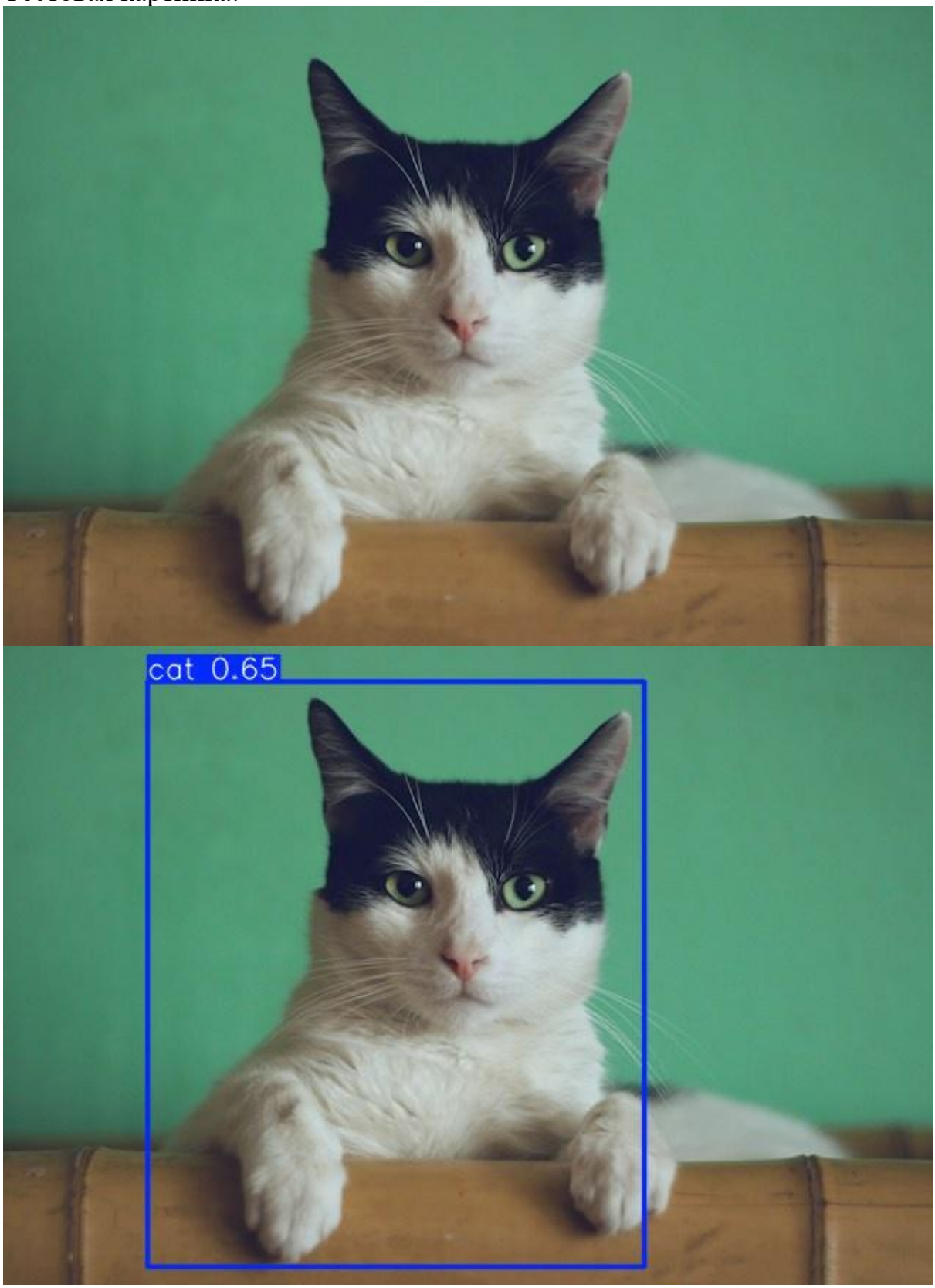
Validating C:\Lesha\4курс задания\ОИВИС\lab3\runs\train\weights\best.pt...
Ultralytics 8.4.211  Python-3.13.3 torch-2.15.0+cpu CPU (11th Gen Intel Core i5-11400H @ 2.70GHz)
YOLO11s summary (fused): 100 layers, 9,413,187 parameters, 0 gradients, 21.4 GFLOPs
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 15/15 1.2s/it 17.4s
                   all        232        235      0.917      0.911      0.945      0.568
Speed: 1.8ms preprocess, 69.3ms inference, 0.0ms loss, 0.5ms postprocess per image
Results saved to C:\Lesha\4курс задания\ОИВИС\lab3\runs\train
Ultralytics 8.4.211  Python-3.13.3 torch-2.15.0+cpu CPU (11th Gen Intel Core i5-11400H @ 2.70GHz)
YOLO11s summary (fused): 100 layers, 9,413,187 parameters, 0 gradients, 21.4 GFLOPs
WARNING val: Slow image access detected (ping: 0.1±0.1 ms, read: 4.0±1.0 MB/s, size: 24.6 KB). Use local storage instead of remote/mounted storage for bette
r performance. See https://docs.ultralytics.com/guides/model-training-tips
val: Scanning C:\Lesha\4курс задания\ОИВИС\lab3\dataset\test\labels... 116 images, 0 backgrounds, 0 corrupt: 100% ━━━━━━━━━━━━ 116/116 531.5it/s 0.2s
val: New cache created: C:\Lesha\4курс задания\ОИВИС\lab3\dataset\test\labels.cache
                 Class     Images  Instances      Box(P          R      mAP50  mAP50-95): 100% ━━━━━━━━━━━━ 8/8 1.1s/it 9.0s
                   all        116        116      0.872       0.94       0.96      0.592
Speed: 2.2ms preprocess, 70.4ms inference, 0.0ms loss, 0.6ms postprocess per image
Results saved to C:\Lesha\4курс задания\ОИВИС\lab3\runs\detect\val
mAP50=0.9598  mAP50-95=0.5915

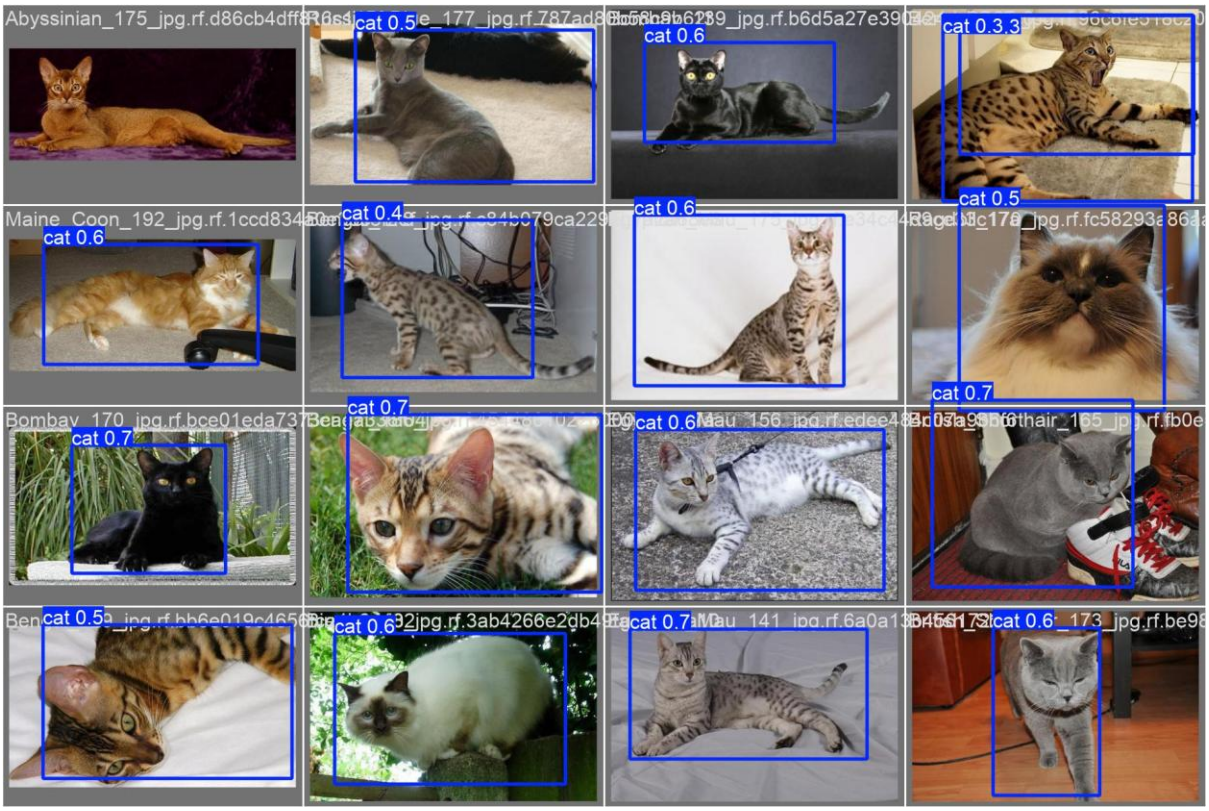
0: 640x640 1 cat, 87.2ms
1: 640x640 1 cat, 87.2ms
2: 640x640 1 cat, 87.2ms
3: 640x640 1 cat, 87.2ms
4: 640x640 1 cat, 87.2ms
Speed: 2.0ms preprocess, 87.2ms inference, 0.5ms postprocess per image at shape (1, 3, 640, 640)
Results saved to C:\Lesha\4курс задания\ОИВИС\lab3\runs\vis
Готово! Результаты сохранены в папке runs/vis
```

# График ошибок:



Тестовая картинка:






**Вывод:** осуществил обучение нейросетевого детектора для решения задачи обнаружения заданных объектов